

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 15

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Recap of Previous Lecture



- ✓ Topic Basic concepts associated with group
- ✓ Topic Finite groups
- ✓ Topic Addition modulo m
- ✓ Topic Multiplication modulo m

Topics to be Covered



✓ **Topic**

Order of an element in a group

✓ **Topic**

Sub-group





Topic : Special Sets

- ✓ N - Set of all natural numbers
- ✓ Z = Set of all integers
- ✓ R - Set of all real numbers
- ✓ Q = Set of all rational numbers
- ✓ Q^* = Set of all non-zero rational numbers



Topic : Algebraic Structure

/ Groupoid

Let 'S' be any non-empty set, and '*' is some binary operation,

then set S w.r.t. binary opⁿ '*' is called a groupoid if and only if

$$a * b \in S, \forall a, b \in S$$

↳ it is called Closure property.

i.e., Set S is Closed w.r.t. binary opⁿ '*'

Set 'S' w.r.t. binary opⁿ '*'
Can be denoted by $(S, *)$

	Algebraic Structure (Groupoid)	Semi-Group	Monoid	Group	Abelian Group
<u>N</u>	$(N, +)$	✓			
	$(N, \cdot)$	✓			
	$(N, -)$	✗			
	$(N, \div)$	✗			
<u>Z</u>	$(Z, +)$	✓			
	$(Z, \cdot)$	✓			
	$(Z, -)$	✓			
	$(Z, \div)$	✗			
<u>Q</u>	$(Q, +)$	✓			
	$(Q, \cdot)$	✓			
	$(Q, -)$	✓			
	$(Q, \div)$	✗ $0 \in Q, \neq 0 \div 0$ not defined			
<u>Q*</u>	$(Q^*, +)$	✗ $\frac{p}{q} + (\frac{-p}{q}) = 0 \notin Q^*$			
	$(Q^*, \cdot)$	✓			
	$(Q^*, -)$	✗ $\frac{p}{q} - (\frac{p}{q}) = 0 \notin Q^*$			
	$(Q^*, \div)$	✓			

$$(2-5) = -3 \notin N$$

$$2 \div 3 \notin Z$$

$$0 \in Q, \neq 0 \div 0 \text{ not defined}$$

$$\frac{p}{q} + (\frac{-p}{q}) = 0 \notin Q^*$$

$$\frac{p}{q} - (\frac{p}{q}) = 0 \notin Q^*$$



Topic : Semi-group

A groupoid $(S, *)$ is called a semi-group if and only if following property holds true

$$(a * b) * c = a * (b * c), \forall a, b, c \in S$$

↳ i.e, operation must be associative

	Algebraic Structure (Groupoid)	Semi-Group	Monoid	Group	Abelian Group
$\mathbb{N}$	$(\mathbb{N}, +)$	✓			
	$(\mathbb{N}, \cdot)$	✓			
	$(\mathbb{N}, -)$	✗			
	$(\mathbb{N}, \div)$	✗			
$\mathbb{Z}$	$(\mathbb{Z}, +)$	✓			
	$(\mathbb{Z}, \cdot)$	✓			
	$(\mathbb{Z}, -)$	✓	✗		
	$(\mathbb{Z}, \div)$	✗	✗		
$\mathbb{Q}$	$(\mathbb{Q}, +)$	✓			
	$(\mathbb{Q}, \cdot)$	✓			
	$(\mathbb{Q}, -)$	✓	✗		
	$(\mathbb{Q}, \div)$	✗	✗		
$\mathbb{Q}^*$	$(\mathbb{Q}^*, +)$	✗	✗		
	$(\mathbb{Q}^*, \cdot)$	✓	✓		
	$(\mathbb{Q}^*, -)$	✗	✗		
	$(\mathbb{Q}^*, \div)$	✓	✗		



Topic : Monoid



A semi-group $(S, *)$ is called a Monoid if and only if there exists an element " $e \in S$ "

s.t

$$a * e = a, \forall a \in S$$

and

$$e * a = a, \forall a \in S$$

↳ Such an element ' e ' is called identity element w.r.t. binary opⁿ ' $*$ '

	Algebraic Structure (Groupoid)	Semi-Group	Monoid	Group	Abelian Group
$\mathbb{N}$	$(\mathbb{N}, +)$	✓	$0 \notin \mathbb{N}$ ✗		
$\mathbb{N}$	$(\mathbb{N}, \cdot)$	✓	$1 \in \mathbb{N}$ ✓		
$\mathbb{N}$	$(\mathbb{N}, -)$	✗	✗		
$\mathbb{N}$	$(\mathbb{N}, \div)$	✗	✗		
$\mathbb{Z}$	$(\mathbb{Z}, +)$	✓	$0 \in \mathbb{Z}$, ✓		
$\mathbb{Z}$	$(\mathbb{Z}, \cdot)$	✓	$1 \in \mathbb{Z}$, ✓		
$\mathbb{Z}$	$(\mathbb{Z}, -)$	✓	✗		
$\mathbb{Z}$	$(\mathbb{Z}, \div)$	✗	✗		
$\mathbb{Q}$	$(\mathbb{Q}, +)$	✓	$0 \in \mathbb{Q}$, ✓		
$\mathbb{Q}$	$(\mathbb{Q}, \cdot)$	✓	$1 \in \mathbb{Q}$, ✓		
$\mathbb{Q}$	$(\mathbb{Q}, -)$	✓	✗		
$\mathbb{Q}$	$(\mathbb{Q}, \div)$	✗	✗		
$\mathbb{Q}^*$	$(\mathbb{Q}^*, +)$	✗	✗		
$\mathbb{Q}^*$	$(\mathbb{Q}^*, \cdot)$	✓	$1 \in \mathbb{Q}^*$, ✓		
$\mathbb{Q}^*$	$(\mathbb{Q}^*, -)$	✗	✗		
$\mathbb{Q}^*$	$(\mathbb{Q}^*, \div)$	✓	✗		

$\mathbb{N}$
 $= \{1, 2, \dots\}$

$\mathbb{Z}$

$\mathbb{Q}$

$\mathbb{Q}^*$



Topic : Group

A monoid $(S, *)$ is called a group,
if and only if. for every element $a \in S$
there exists some element $b \in S$,
Such that

$$\begin{aligned} a * b &= e \text{ (identity)} \\ \text{(and)} \\ b * a &= e \end{aligned}$$

→ Here 'a' & 'b' are called
inverse of each other
i.e. for a group, inverse of
every element of the set
must exist within the set itself.



	Algebraic Structure (Groupoid)	Semi-Group	Monoid	Group	Abelian Group
N	$(N, +)$	✓	$0 \notin N$ ✗	✗	
	$(N, \cdot)$	✓	$1 \in N$ ✓	✗	inverse does not exist for any element except for '1'
	$(N, -)$	✗	✗	✗	
	$(N, \div)$	✗	✗	✗	
Z	$(Z, +)$	✓	$0 \in Z$ ✓	✓	inverse does not exist for any element except for '1' & '-1'
	$(Z, \cdot)$	✓	$1 \in Z$ ✓	✗	
	$(Z, -)$	✓	✗	✗	
	$(Z, \div)$	✗	✗	✗	
Q	$(Q, +)$	✓	$0 \in Q$ ✓	✓	A set containing '0' can never be a group w.r.t. multiplication.
	$(Q, \cdot)$	✓	$1 \in Q$ ✓	✗	
	$(Q, -)$	✓	✗	✗	
	$(Q, \div)$	✗	✗	✗	
Q*	$(Q^*, +)$	✗	✗	✗	
	$(Q^*, \cdot)$	✓	$1 \in Q^*$ ✓	✓	
	$(Q^*, -)$	✗	✗	✗	
	$(Q^*, \div)$	✓	✗	✗	



Topic : Abelian group / Commutative group

A group $(S, *)$ is called an Abelian group
if and only if,

$$a * b = b * a, \quad \forall a, b \in S$$

→ i.e, opⁿ '*' must be commutative
on the elements of set S.

	Algebraic Structure (Groupoid)	Semi-Group	Monoid	Group	Abelian Group
$\mathbb{N}$	$(\mathbb{N}, +)$	✓	$0 \notin \mathbb{N}$ ✗	✗	✗
	$(\mathbb{N}, \cdot)$	✓	$1 \in \mathbb{N}$ ✓	✗	✗
	$(\mathbb{N}, -)$	✗	✗	✗	✗
	$(\mathbb{N}, \div)$	✗	✗	✗	✗
$\mathbb{Z}$	$(\mathbb{Z}, +)$	✓	$0 \in \mathbb{Z}$, ✓	✓	✓
	$(\mathbb{Z}, \cdot)$	✓	$1 \in \mathbb{Z}$, ✓	✗	✗
	$(\mathbb{Z}, -)$	✓	✗	✗	✗
	$(\mathbb{Z}, \div)$	✗	✗	✗	✗
$\mathbb{Q}$	$(\mathbb{Q}, +)$	✓	$0 \in \mathbb{Q}$, ✓	✓	✓
	$(\mathbb{Q}, \cdot)$	✓	$1 \in \mathbb{Q}$, ✓	✗, $0 \in \mathbb{Q}$,	✗
	$(\mathbb{Q}, -)$	✓	✗	✗	✗
	$(\mathbb{Q}, \div)$	✗	✗	✗	✗
$\mathbb{Q}^*$	$(\mathbb{Q}^*, +)$	✗	✗	✗	✗
	$(\mathbb{Q}^*, \cdot)$	✓	$1 \in \mathbb{Q}^*$, ✓	✓	✓
	$(\mathbb{Q}^*, -)$	✗	✗	✗	✗
	$(\mathbb{Q}^*, \div)$	✓	✗	✗	✗

Note

- ✓ ① Inverse of identity element will be identity element itself.
- ✓ ② Identity element (if exists) in the set, then it will be unique w.r.t. given binary opⁿ '*'.
- ✓ ③ In a group inverse of every element of the set exists within the set itself.
and if $\text{inv}(a) = b$ then $\text{inv}(b) = a$ $\text{i.e. } (a^{-1})^{-1} = a$
 $a^{-1} = b$ $b^{-1} = a$
- ✓ ④ In a group $(G, *)$, $(a * b)^{-1} = b^{-1} * a^{-1}$ always holds true $\forall a, b \in G$

Groupoid	Semi-group	Monoid	Group	Abelian group
① Closed	① Closed ② Associative	① Closed ② Associative ③ Identity element must be present within the set	① Closed ② Associative ③ Identity element must be present within the set ④ Inverse of every element of the set must be present within the set	① Closed ② Associative ③ Identity element must be present within the set ④ Inverse of every element of the set must be present within the set ⑤ Commutative

Note:-

Let 'S' is any non-empty set.



Set 'S' is a group w.r.t. binary opn '*'

if and only if,

①

$$a * b^{-1} \in S, \forall a, b \in S$$

②

'*' is associative

and

Proof on next slide

This statement is
sufficient to prove that,

① 'S' is closed w.r.t. '*'

② Identity w.r.t. '*' $\in S$

③ $a^{-1} \in S, \forall a \in S$

* Let $a * b^{-1} \in S, \forall a, b \in S.$

① Identity:

Let $a \in S$

$\therefore a, a \in S$

We know $\forall a, b \in S, a * b^{-1} \in S$

$\therefore$ if $a, a \in S$ then $a * a^{-1} \in S$

$e \in S$

i.e., identity element will be present in set S .

② Inverse:

* If identity element 'e' is the only element in the set, then we know $\text{inv}(e) = e$.

$\therefore$ Inverse exist for Every element of the set.

* If some other element $a \in S$, such that $a \neq e$

then $e, a \in S$ { We already Proved that $e \in S$ }

$\therefore e * (a^{-1}) \in S$

$a^{-1} \in S$

i.e., if $a \in S$ then $a^{-1} \in S$

③ Closed.

Let $a, b \in S$

We just proved that if $a, b \in S$, then $a^{-1}, b^{-1} \in S$

i.e., if $a, b \in S$

then $a, b^{-1} \in S$ is also true

$\therefore a * (b^{-1})^{-1} \in S$

if $a, b \in S$

then $a * b \in S$

Hence Closed

Finite Group

- A group $(G, *)$ is called a finite group if and only if set ' G ' is a non-empty finite set.
- If $(G, *)$ is a finite group, then Order of group $(G, *)$ is same as Cardinality of set G !

i.e.

$$O(G) = |G|$$

↑
Order of
group G

Notes:

- ① $\{0\}$ is a finite group of order = 1, w.r.t. addition
- ② $\{1\}$ is a finite group of order = 1, w.r.t. multiplication.
- ③ $\{-1, 1\}$ is a finite group of order = 2, w.r.t. multiplication

The above three groups are the only finite groups
of real numbers w.r.t. addition and/or multiplication.

Note:

If element ' e ' is an identity element w.r.t. binary opn ' $*$ ', then $\{e\}$ is always a finite group of 'order=1' w.r.t. opn ' $*$ '.

Note:

In a finite group of order=2,
Every element is inverse of itself.

eg: $1, \omega$ & ω^2 are Cube roots of unity where $\boxed{\omega^3=1}$

The set $\{1, \omega, \omega^2\}$ form a finite group of order = 3

w.r.t. binary opn multiplication.

To find inverse we can use the concept of Composition table

opn is multiplication

① We know $\omega^3=1$,
∴ Set is closed w.r.t. multiplication

② Multiplication is associative

③ Identity w.r.t. multiplication is '1'
And $1 \in$ Set.

④ Inverse, $\left. \begin{array}{l} \text{inv}(1)=1 \\ \text{inv}(\omega)=\omega^2 \\ \text{inv}(\omega^2)=\omega \end{array} \right\} \begin{array}{l} \text{inverse} \\ \text{exist for} \\ \text{Every element} \end{array}$

	1	ω	ω^2
1	1 = e	ω	ω^2
ω	ω	ω^2	$\omega^3 = 1 = e$
ω^2	ω^2	$\omega^3 = 1 = e$	$\omega^4 = \omega^3 \cdot \omega = 1 \cdot \omega = \omega$

Annotations on the table:
 $\text{inv}(1) = 1$ (green arrow from 1 to 1)
 $\text{inv}(\omega) = \omega^2$ (red arrow from ω to ω^2)
 $\text{inv}(\omega^2) = \omega$ (blue arrow from ω^2 to ω)

eg:- $1, -1, i$ & $-i$ are four roots of Unity.

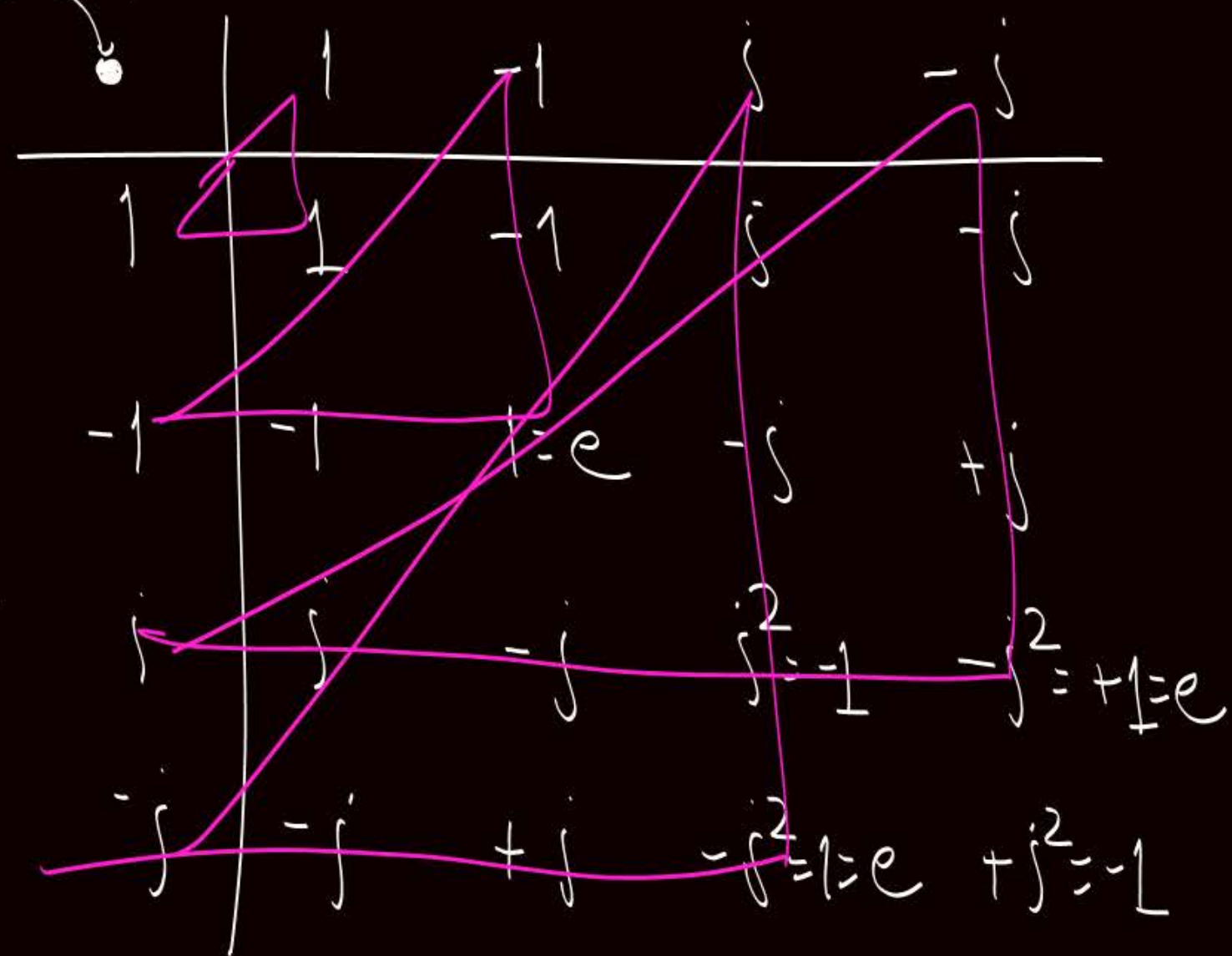
The set $\{1, -1, i, -i\}$ form a finite group of order = 4,
w.r.t. binary opⁿ multiplication.

- ✓ ① $i^2 = -1$, ∴ Closed
- ✓ ② Multiplication is associative
- ✓ ③ Identity, i.e., $1 \in$ Set
- ④ Inverse,

$$\begin{aligned} \text{inv}(1) &= 1 \\ \text{inv}(-1) &= -1 \\ \text{inv}(i) &= -i \\ \text{inv}(-i) &= i \end{aligned}$$

} inverse exist
for every element.

multiplication



* Note:- The set of 'n' roots of unity will form a finite group of order = n w.r.t. multiplication.



Topic : Addition modulo ' m ' $\oplus_m$

- Let ' m ' is a fixed positive integer. *it gives remainder*
- for any two non-negative integers a & b

$$a \oplus_m b = \begin{cases} a+b & \text{if } (a+b) < m \\ \pi & \text{otherwise} \end{cases}$$

$\equiv (a+b) \% m$

eg.

$$2 \oplus_8 3 = 5$$

$$4 \oplus_8 5 = 3$$

π is the remainder obtained on dividing $(a+b)$ by m



Topic : Multiplication modulo 'm' $\otimes_m$

- Let m is a fixed ^{it gives remainder} positive integer

• For any two non-negative integer a & b

$$a \otimes_m b = \begin{cases} (a \cdot b) & \text{if } (a \cdot b) < m \end{cases}$$

$$(a \cdot b) \% m$$

r

(remainder)

Otherwise

eg.

$$2 \otimes_7 3 = 6$$

$$4 \otimes_7 5 = 6$$



Topic : NOTE

① The set $\{0, 1, 2, \dots, (n-1)\}$ form a finite group of order = n w.r.t. binary opⁿ $\oplus_n$

eg: $\{0, 1, 2, 3, 4\}$ is a finite group of order = 5 w.r.t $\oplus_5$

① Closed ✓

② $\oplus_n$ is associative for any positive integer 'n'

③ Identity w.r.t. $\oplus_n = 0 \in \text{Set}$ ✓

④ inverse:-

$$\text{inv}(0) = 0, \quad \text{inv}(1) = 4$$

$$\text{inv}(2) = 3, \quad \text{inv}(3) = 2, \quad \text{inv}(4) = 1$$

$$\begin{aligned} \text{inv}(0) &= 0 \\ \text{inv}(x) &= (n-x) \end{aligned}$$

$$\forall x \neq 0$$

Note:-

let 'n' is given

ie, x & n are Co-prime
or "relatively prime"

$$\text{let } A = \{x \mid x \in \mathbb{N} \text{ and } x < n \text{ and } \text{GCD}(x, n) = 1\}$$

$\mathbb{N} = \{1, 2, 3, \dots\}$

Here, A is a set of all natural numbers
that are less than 'n' and Co-prime to 'n'

eg. let $n = 15$, then $A = ?$

$$A = \{1, 2, 4, 7, 8, 11, 13, 14\}$$



Topic : NOTE

- ② The set of all natural numbers that are less than ' n ' and Co-prime to ' n ' definitely form a group w.r.t. multiplication modulo ' n ' { i.e; $\otimes_n$ }

eg. let $n=15$, then
 $\{1, 2, 4, 7, 8, 11, 13, 14\}$ is group of order = 8 w.r.t.
binary opⁿ $\otimes_{15}$

① Closed ✓

② $\otimes_n$ is associative
for any +ve integer 'n'

③ Identity w.r.t $\otimes_n = 1$

④ Inverse: -

$$\text{inv}(1) = 1 \quad 1 \otimes_{15} 1 = 1$$

$$\text{inv}(2) = 8 \quad 2 \otimes_{15} 8 = 1$$

$$\text{inv}(4) = 4 \quad 4 \otimes_{15} 4 = 1$$

$$\text{inv}(7) = 13 \quad 7 \otimes_{15} 13 = 1$$

$$\text{inv}(8) = 2$$

$$\text{inv}(11) = 11 \quad 11 \otimes_{15} 11 = 1$$

$$\text{inv}(13) = 7$$

$$\text{inv}(14) = 14 \quad 14 \otimes_{15} 14 = 1$$

eg. Let $n=8$, then

$\{1, 3, 5, 7\}$ is a finite group of order = 4 w.r.t. $\otimes_8$

① Closed ✓

② Associative ✓

③ Identity $\in S$

④ Inverse

$$\text{inv}(1) = 1$$

$$\text{inv}(3) = 3$$

$$\text{inv}(5) = 5$$

$$\text{inv}(7) = 7$$

$$1 \otimes_8 1 = 1$$

$$3 \otimes_8 3 = 1$$

$$5 \otimes_8 5 = 1$$

$$7 \otimes_8 7 = 1$$

In this group every element is inverse of itself

eg. ✓ ① $\{1, 3, 5, 7\}$ is a group of order = 4 w.r.t. $\otimes_8$

$\{1\}$ is a finite group of order = 1 w.r.t. $\otimes_8$

Note: $\{1, 3\}$ is a finite group of order = 2 w.r.t. $\otimes_8$

$\{1, 5\}$ is a finite group of order = 2 w.r.t. $\otimes_8$

$\{1, 7\}$ is a finite group of order = 2 w.r.t. $\otimes_8$

← Some subsets of original set may also be the group w.r.t. same binary opⁿ

→ But all subsets need not be the group w.r.t. same binary opⁿ.

eg. $\{3, 5\}$ is not closed w.r.t. $\otimes_8 \therefore$ it is not a group.

eg. $\{1, 3, 5\}$ is not closed w.r.t. $\otimes_8 \therefore$ it is not a group.

eg. We know $\{0, 1, 2, 3\}$ is a group w.r.t. $\oplus_4$

Here $\{0\}$ is also a group w.r.t. $\oplus_4$ ✓

* $\{0, 2\}$ is also a group w.r.t. $\oplus_4$

* $\{0, 1\}$ is not closed w.r.t. $\oplus_4 \therefore$ not a group

Note:- We know,

The set of all natural numbers that are less than n and co-prime to n form a group w.r.t. binary opⁿ $\otimes_n$

o.o if ' p ' is a prime number then the set $\{1, 2, 3, \dots, (p-1)\}$ will form a finite group of order $= (p-1)$ w.r.t. binary opⁿ $\otimes_p$ ✓

eg. The set $\{1, 2, 3, 4, 5, 6\}$ is a group of order $= 6$ w.r.t. opⁿ $\otimes_7$

HoWo

#Q. Let $A = \{0, \pm 2, \pm 4, \pm 6, \dots\}$
 $B = \{0 \pm 1, \pm 3, \pm 5, \dots\}$

Which of the following is not a semi-group

Closed

Associative

A $(A, +)$

B $(A, \cdot)$

C $(B, +)$ (Not Closed)

D $(B, \cdot)$

How.

NULL string (ϵ) is also part of Σ^*

#Q. Consider the set Σ^* of all strings over the alphabet $\Sigma = \{0, 1\}$. Σ^* with the concatenation operator for strings

A

~~Not a semigroup~~

B

Semi group but not a monoid

C

Monoid but not a group.

D

A group

① Closed ✓

② $(s_1 \odot s_2) \odot s_3 = s_1 \odot (s_2 \odot s_3) = \underline{s_1 s_2 s_3}$
 $\therefore$ Associative ✓

③ Identity w.r.t. Concatenation is ' ϵ '
 $s_1 \odot \epsilon = s_1, \forall s_1 \in \Sigma^* \nmid \epsilon \in \Sigma^*$

④ Inverse does not exist for any string
 Except for identity (ϵ)

H.W

#Q. Let A be the set of all $\overbrace{\text{non-singular matrices}}^{(\det) \neq 0}$ of order $n \times n$ over real number and let $*$ be the matrix multiplication operation. Then

$$\textcircled{1} \quad A_{n \times n} \cdot B_{n \times n} = (AB)_{n \times n}$$

if $|A| \neq 0$ & $|B| \neq 0$
then $|AB| \neq 0$
is closed

A A is closed under $*$ but $\langle A, * \rangle$ is not a semigroup

B $\langle A, * \rangle$ is a semigroup but not a monoid.

C $\langle A, * \rangle$ is a monoid but not a group.

D $\langle A, * \rangle$ is a group but not an abelian group.

$\textcircled{2}$ Matrix multiplication is associative

$\textcircled{3}$ Identity w.r.t. Matrix Multiplication is $I_{n \times n}$ (Identity Matrix)

H.W.

Finite Set

#Q. Let S be any finite set, and $F(S)$ is defined as set of all function on set S. Then $F(S)$ with respect to function composition operation (ie., $\circ$) is.

A

Not a semigroup

B

Semi group but not a monoid

C

Monoid but not a group.

D

A group

① let $f_1, f_2 \in F(S)$ then $f_1: S \rightarrow S$ & $f_2: S \rightarrow S$
 $\therefore f_1 \circ f_2: S \rightarrow S$ & $f_2 \circ f_1: S \rightarrow S$
 $\therefore$ Closed

② Function composition is Associative
 $(f \circ g) \circ h = f \circ (g \circ h)$

③ Identity function on set S i.e. $I_S \in F(S)$
 $\& \text{it } f \circ I_S = I_S \circ f = f, \forall f \in F(S)$
 $\therefore$ Identity exist.

④ Every function on set S need not be bijective
 $\therefore$ Inverse need not exist
 (Not a group)

H.W.

#Q. Let Z is the set of all integers. The binary operation $*$ is defined as $a*b = \max(a, b)$ then the structure $(Z, *)$ is

A

Not a semigroup

B

Semi group but not a monoid

C

Monoid but not a group.

D

A group

①

$a*b = \max(a, b) = a \text{ or } b$
 $\text{If } a, b \in Z \text{ then } \max(a, b) \in Z \therefore \text{closed}$

②

$\max(a, \max(b, c)) = \max(\max(a, b), c)$

Result will be max of a, b, c
 $\therefore$ Associative

③

Identity element will be the smallest integer

$\downarrow$
it does not exist.
 $\therefore$ Not a Monoid.

H.W.

#Q. Let Q^+ be the set of all positive rational numbers. The binary operation $*$ is

defined as $a * b = \frac{ab}{3} \forall a, b \in Q^+$ If $(Q^+, *)$ is a group then find

- (i) identity element of the group
- (ii) inverse of any element $a, \forall a \in \text{Group}$

$$a * e = a$$

$$\frac{a \cdot e}{3} = a$$

$$e = 3$$

$$a * a^{-1} = e$$

$$\frac{a \cdot a^{-1}}{3} = 3$$

$$a^{-1} = \frac{9}{a}$$

H.W.

#Q. Which of the following statement is/are not true.

A $\{0, \pm 2k, \pm 4k, \pm 6k, \dots\}$ is a group with respect to addition where k is any fixed positive integer (True)

B $\{x \mid x \text{ is real number and } 0 < x \leq 1\}$ is a group with respect to multiplication
 (Closed Associative Identity) Inv(1)=1, & inv does not exist for any other element

C $\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication (True)

Let $n_1, n_2 \in \text{integer}$ then $n_1 + n_2 \in \text{integer}$
 if $n_1, n_2 \in \text{int}$ then $2^{n_1}, 2^{n_2} \in \text{Set}$, & $2^{n_1} \cdot 2^{n_2} = 2^{n_1+n_2}$ } Closed ✓
 it is also int belongs to set
 $0 = \text{int}$
 $2^0 \in \text{Set}$
 $1 \in \text{Set} \therefore \text{identity exist}$

H.W.

H.W. Q: Which of the following is/are not true?

✓ (a) $\{1, 2, 3, 4, 5\}$ is a group w.r.t. $\oplus_6$
 $2 \oplus_6 4 = 0 \notin \text{Set} \therefore \text{Not Closed}$

$2 \oplus_6 4 = 0 \notin \text{Set} \therefore \text{Not Closed}$

⑥ $\{1, 2, 3, 4, 5\}$ is a group w.r.t. $\otimes_6$
 $2 \otimes_6 3 = 0 \notin \text{Set} \therefore \text{Not closed}$

$$2 \otimes 3 = 0 \notin \text{Set} \therefore \text{Not closed}$$

© $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$ (True)

✓ (d) $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\otimes_5$

Closed ✓ Associative ✓ Identity exist ✓

$0 \in \text{Set}$
 $\& \text{inv}(0) = \text{does not exist}$



Topic : Order of an element in a group $(G, *)$

- Let $(G, *)$ be a finite group such that order of group $(G, *)$ is $|G|$.
- For an element $a \in G$, Order of element 'a' can be denoted by $O(a)$, and it is defined as

ie,
Can not be '0'

the least positive integer 'n' such that

$$(a)^n = \text{identity}(e) \text{ element of group } (G, *)$$

binary opⁿ
w.r.t. group

$$\underbrace{a * a * a \dots * a}_{\text{'n' times a}}$$

Note: Let $(G, *)$ is finite group and 'e' is the identity element of group $(G, *)$.

$$(e)^1 = e(\text{identity})$$

$$\therefore \boxed{O(e) = 1}$$

Order of identity element is always '1'

eg. We know $\{1, -1\}$ is a group of order = 2 w.r.t. 'Multiplication'
Find order of every element of group.

$$O(G) = 2$$

• $1 = \text{identity}$. $\therefore O(1) = 1$

• $(-1)^1 = -1 \neq e$

$$(-1)^2 = 1 = e$$

$$\therefore O(-1) = 2$$

eg. $\{1, \omega, \omega^2\}$ is a group w.r.t. multiplication such that $O(G) = 3$

Find order of every element of group

* $1 = \text{identity} \quad \therefore \boxed{O(1) = 1}$

* $(\omega)^1 = \omega$
 $(\omega)^2 = \omega^2$
 $(\omega)^3 = \omega^3 = 1 = \text{identity} \quad \therefore \boxed{O(\omega) = 3}$

* $(\omega^2)^1 = \omega^2$
 $(\omega^2)^2 = \omega^4 = \omega$
 $(\omega^2)^3 = \omega^6 = \omega^3 \omega^3 = 1 = \text{identity} \quad \therefore \boxed{O(\omega^2) = 3}$

eg $\{1, -1, i, -i\}$ is a group w.r.t multiplication
Such that $O(G) = 4$
Find order of every element of group.

$\rightarrow 1 = \text{identity} \quad \therefore O(1) = 1$

$\rightarrow (-1)^1 = -1$
 $(-1)^2 = 1 = \text{identity} \quad \therefore O(-1) = 2$

$\rightarrow (i)^1 = i$
 $(i)^2 = -1$
 $(i)^3 = -i$
 $(i)^4 = 1 = \text{identity} \quad \therefore O(i) = 4$

$$(-i)^1 = -i$$

$$(-i)^2 = -1$$

$$(-i)^3 = i$$

$$(-i)^4 = 1 = \text{identity} \quad \therefore O(-i) = 4$$



Topic : Properties w.r.t. Finite Group

① Order of identity element is always '1'

② Let $(G, *)$ be a finite group, then order of every element of group $(G, *)$ will be less than or equal to the order of group $(G, *)$, and order of every element of group $(G, *)$ divides the order of group $(G, *)$

It is
Lagrange's
Theorem

Let $(G, *)$ be a group of order $|G|$
then for all $a \in G$
 $O(a)$ divides $|G|$



Topic : Properties w.r.t. Finite Group

③ Order of an element $a \in$ group $(G, *)$ is Exactly same as order of its inverse

i.e.

$$O(a) = O(a^{-1})$$

④ Order of an element 'a' of group $(G, *)$ can be '1' if and only if 'a' is the identity element w.r.t. group $(G, *)$

⑤ For an element 'a' of group $(G, *)$

if $O(a) = 2$ (i.e. $a * a = e$ (identity)), then element 'a' is inverse of itself
i.e. if $O(a) = 2$, then $a^{-1} = a$

Q: We know $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$.
Find the order of every element of the group.

→ $0 = \text{identity}$

∴ $\boxed{O(0) = 1}$

→ $(1)^1 = 1$

$(1)^2 = 1 \oplus_5 1 = 2$

$(1)^3 = 1 \oplus_5 1 \oplus_5 1$

$= (1 \oplus_5 1) \oplus_5 1$

$= (1)^2 \oplus_5 1$

$= 2 \oplus_5 1 = 3$

$(1)^4 = (1)^3 \oplus_5 1 = 3 \oplus_5 1 = 4$

$(1)^5 = (1)^4 \oplus_5 1 = 4 \oplus_5 1 = 0 = \text{identity}$

∴ $\text{inv}(1) = 4$

∴ $\boxed{O(1) = 5}$

∴ $\boxed{O(4) = O(1) = 5}$

if $O(a) = n$
then $\text{inv}(a) = (a)^{n-1}$

$(2)^1 = 2$

$(2)^2 = 2 \oplus_5 2 = 4$

$(2)^3 = (2)^2 \oplus_5 2 = 1$

$(2)^4 = (2)^3 \oplus_5 2 = 3$

$(2)^5 = (2)^4 \oplus_5 2 = 0 = \text{identity}$

∴ $\boxed{O(2) = 5}$

∴ $\text{inv}(2) = 3$

∴ $\boxed{O(3) = O(2) = 5}$

Q: We know $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$.
Find the order of every element of the group.

order of group = 5 $\leftarrow$ it is a prime no.

We know order of every element must divide the order of group

$\therefore$ Order of Every Element must divide 5 (Prime no.)

$\rightarrow$ 5 can be divided only by '1' or '5'

$\therefore O(\text{element}) = 1$

$\downarrow$
it can be true
only for identity
element

$\therefore \boxed{O(0) = 1}$

or $O(\text{element}) = 5$

$\downarrow$
Order = 5 for Every non-identity
element

$\therefore \boxed{O(1) = O(2) = O(3) = O(4) = 5}$

Sub-group

Sub-group

Let $(G, *)$ be a group, a non-empty subset H of set G is called a subgroup of group $(G, *)$ only if $(H, *)$ is a group.

Sub-group

Note:- ① Every set is a subset of itself, $\therefore$ Every group is a subgroup of itself.

(2) If 'e' is the identity element of group $(G, *)$
then $(\{e\}, *)$ is always a subgroup of group $(G, *)$
subset of set G

* If $(G, *)$ is group such that 'e' is the identity element, then

group $(G, *)$ itself and group $(\{e\}, *)$ are always sub-group of group $(G, *)$ and they are called "trivial subgroup" of group $(G, *)$

Any other sub-group of group $(G, *)$ is called Proper-sub-group of group $(G, *)$

eg. We know $\{1, -1, i, -i\}$ is a group w.r.t. multiplication

- trivial {
- ① $\{1, -1, i, -i\}$ is a trivial sub-group of above group
 - ② $\{1\}$ is also a trivial sub-group of above group
- Proper sub-group {
- ③ $\{1, -1\}$ is a proper sub-group of above group

No other sub-group is possible for given group.

eg:

We know $\{1, 3, 5, 7\}$ is a group of order=4 w.r.t. $\otimes_8$

$\{1, 3, 5, 7\}$ & $\{1\}$ are trivial sub-group of above group w.r.t. $\otimes_8$

Following are the proper sub-groups of above group

- ① $\{1, 3\}$ is a group of order=2 w.r.t. $\otimes_8$
- ② $\{1, 5\}$ is a group of order=2 w.r.t. $\otimes_8$
- ③ $\{1, 7\}$ is a group of order=2 w.r.t. $\otimes_8$

Properties w.r.t. Subgroup

- ① Let $(G, *)$ is a group, and H is any non-empty subset of set G , then
- $(H, *)$ is a subgroup of group $(G, *)$
- if and only if, $a * b^{-1} \in H, \forall a, b \in H$

Properties w.r.t. Subgroup

② Let $(G, *)$ be a group and $(H, *)$ be a subgroup of group $(G, *)$ such that $|G| = \text{order of group } (G, *)$ and $|H| = \text{order of subgroup } (H, *)$ then $|H|$ divides $|G|$

It is also Lagrange's Theorem

i.e., Order of subgroup will always divide the order of original group.

Q: Let $(G, *)$ be a group of prime order = P ,
How many subgroup exists for group $(G, *)$

$$O(G) = P = \text{prime no.}$$

order of sub-group
must divide the
order of group

∴ Possible orders
of sub-group =

1
↓

in a group
w.r.t.
identity element

or

P
↓

in a sub-group which is original
group itself

Only two sub-groups are possible & Both will be trivial.

Q:- Let $(G, *)$ be a group, and $(H_1, *)$ and $(H_2, *)$ are two subgroups of group $(G, *)$,

then which of the following statements is/are always true

✗ S_1 : $(H_1 \cup H_2, *)$ is also a sub-group of $(G, *)$

✓ S_2 : $(H_1 \cap H_2, *)$ is also a sub-group of $(G, *)$

Counter Example

w.r.t. Statement : $S = \{0, \pm 1, \pm 2, \pm 3, \dots\}$ is a group w.r.t. addition

S_1

• Let $H_1 = \{0, \pm 2, \pm 4, \pm 6, \pm 8, \dots\}$ all multiples of 2

& $H_2 = \{0, \pm 3, \pm 6, \pm 9, \pm 12, \dots\}$ all multiples of 3

We can observe both H_1 & H_2 are group w.r.t addition.

And H_1 & H_2 are subset of set S .

i.e. $(H_1, +)$ & $(H_2, +)$ are subgroup of group $(S, +)$

$$H_1 \cup H_2 = \{0, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 9, \dots\}$$

$2 + 3 = 5 \notin H_1 \cup H_2 \therefore H_1 \cup H_2$ is not closed w.r.t. addition
Hence not a group.

Another
Counter Example
w.r.t. Statement :
 S_1

$\{1, 3, 5, 7\}$ is a group w.r.t. $\otimes_8$

Let, $H_1 = \{1, 3\}$ is a sub-group of above group

& $H_2 = \{1, 5\}$ is a sub-group of above group

$H_1 \cup H_2 = \{1, 3, 5\}$ w.r.t. $\otimes_8$ is not closed

$$3 \otimes_8 5 = 7 \notin \{1, 3, 5\}$$



Topic : Properties w.r.t. Subgroup

- ③ Let $(G, *)$ be a group, and $(H_1, *)$ and $(H_2, *)$ be two subgroups of group $(G, *)$ then $(H_1 \cup H_2, *)$ is a subgroup of group $(G, *)$ if and only if $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$
- $\downarrow$
 $H_1 \cup H_2 = H_2$
and it
is a
Sub group
- $\downarrow$
 $H_1 \cup H_2 = H_1$
and it is
a Sub-group

Proof for
Statement:
 S_2

Let $a, b \in H_1 \cap H_2$

then
 $\downarrow$

$a, b \in H_1$

We know H_1 is a group

$$\therefore a * b^{-1} \in H_1$$

and

$a, b \in H_2$

We know H_2 is a group

$$a * b^{-1} \in H_2$$

Hence $a * b^{-1} \in H_1 \cap H_2$

If $a, b \in H_1 \cap H_2$ then $a * b^{-1} \in H_1 \cap H_2$

$\therefore H_1 \cap H_2$ w.r.t. $*$ will be a group.



Topic : Properties w.r.t. Subgroup

④

Let $(G, *)$ be a group, and $(H_1, *)$ and $(H_2, *)$ be two subgroups of group $(G, *)$, then $(H_1 \cap H_2, *)$ is also a subgroup of group $(G, *)$



2 mins Summary



✓
Topic

Order of an element in a group

✓
Topic

Sub-group

THANK - YOU